A Stochastic Model of Infant Cry–Pause Dynamics for Sleep Onset Prediction

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Abstract

Infant crying follows a cyclical pattern of cry and pause phases that gradually attenuate until sleep onset. We propose a generative stochastic model describing this process as a sequence of autocorrelated cry and pause durations with monotonically decreasing and increasing expectations, respectively. Cry decay and pause growth are modulated by a continuous circadian function, and sleep is defined as a random stopping time when cry duration falls below a physiological threshold. We derive the model from first principles, specify all assumptions explicitly, and simulate sleep-onset distributions using Monte Carlo methods. Given a start time of 21:00, the model predicts a mean sleep onset of approximately **[MEAN TIME]** with a 95% prediction interval of **[LOWER]**–**[UPPER]** minutes. We discuss limitations, ethical considerations, and outline a Bayesian hierarchical extension as future work. All code is implemented in Python and made publicly available.

Keywords: stochastic processes, survival analysis, infant sleep, Monte Carlo simulation, circadian rhythm, generative modeling

1 Introduction

1.1 Motivation

The onset of sleep in infants is preceded by a characteristic pattern of alternating crying and quiet periods. Parents and pediatricians have long observed that crying episodes shorten while pauses lengthen, until the infant eventually falls asleep. Despite the ubiquity of this phenomenon, it has received little attention as a formal statistical process.

Understanding sleep onset dynamics matters for at least three reasons. First, it provides a quantitative baseline against which abnormal crying patterns (e.g., colic, illness) can be compared. Second, it enables parents and clinicians to form realistic expectations about sleep latency. Third, it offers a tractable test-bed for stochastic modeling techniques relevant to behavioral time-series data.

1.2 Contributions

We make the following contributions:

(i) We formalize the cry–pause dynamic as a discrete-time stochastic process with autocorrelation, circadian modulation, and a random stopping time.

(ii) We derive the model from first principles and state every assumption explicitly.

(iii) We implement the model in Python and simulate sleep-onset distributions under realistic parameters.

(iv) We discuss the ethical boundaries of predictive models in caregiving contexts.

(v) We propose a Bayesian hierarchical extension to accommodate real-world data and inter-infant heterogeneity.

1.3 Organisation

Section 3 presents the mathematical model. Section 4 describes the simulation methodology. Section 5 reports results. Section 6 discusses limitations and ethics. Section 7 outlines future work. Section 8 concludes.

2 Related Work

The acoustic and behavioral analysis of infant crying has been studied extensively in developmental psychology [1, 5]. These works characterise cry as a communicative signal, but do not model its temporal dynamics.

Stochastic models for fatigue, learning, and drug decay provide natural analogues. Exponential decay appears in pharmacokinetics [4], while logistic saturation describes bounded accumulation processes [6]. Autocorrelated count processes are well-established in time-series literature [2].

Survival analysis [3] frames the sleep onset as a terminal event; we adopt this framing with a discrete-time hazard-like structure.

3 Mathematical Model

3.1 Setup and Notation

Let $i = 1, 2, \ldots$ index the cry–pause cycle. At each cycle i, we observe:

- C_i: cry duration (seconds),
- P_i: pause duration (seconds).

The process begins at clock time $t_0 \in [0, 24)$ (hours). Sleep onset occurs at the first cycle S such that $C_S \leq \delta$, where δ is a physiological threshold.

3.2 Assumptions

We state the following assumptions explicitly.

A1. Autocorrelation. Cry and pause durations across cycles exhibit first-order autocorrelation.

A2. Monotonicity. $\mathbb{E}[C_i]$ is strictly decreasing in i; $\mathbb{E}[P_i]$ is strictly increasing in i.

A3. Exponential cry decay. $\mathbb{E}[C_i] = C_1 e^{-\alpha(t_0)(i-1)}$ for $\alpha > 0$.

A4. Logistic pause growth. $\mathbb{E}[P_i] = P_1 + \beta(1 - e^{-\gamma(t_0)(i-1)})$ for $\beta, \gamma > 0$.

A5. Truncated Gaussian noise. Deviations are Gaussian, truncated below at zero.

A6. Sleep threshold. Sleep occurs at the first i with $C_i \leq \delta$ (e.g., $\delta = 1\,\text{s}$).

A7. No external intervention. The caregiver does not intervene during the process.

A8. Parameter constancy. Parameters are fixed for a given infant during one session.

A9. Bounded cycles. Simulation runs for at most $N_{\max} = 50$ cycles.

A10. Age restriction. Model is calibrated for infants aged 2–6 months.

A11. Circadian modulation. Sleep pressure peaks around 21:00 (night crying leads to faster sleep).

3.3 Expected Values

For cycle i,

$$\mu_i^C = C_1 \cdot e^{-\alpha(t_0)(i-1)}, \tag{1}$$

$$\mu_i^P = P_1 + \beta \left(1 - e^{-\gamma(t_0)(i-1)}\right). \tag{2}$$

3.4 Circadian Modulation

Let ϕ denote the phase (hour of peak sleep pressure). Then

$$\alpha(t_0) = \alpha_0 + \alpha_1 \cos\left(\frac{2\pi(t_0 - \phi)}{24}\right), \tag{3}$$

$$\gamma(t_0) = \gamma_0 + \gamma_1 \cos\left(\frac{2\pi(t_0 - \phi)}{24}\right), \tag{4}$$

with $\alpha_0 > \alpha_1 \geq 0$ and $\gamma_0 > \gamma_1 \geq 0$.

3.5 Autocorrelated Latent States

Let $\tilde{C}_i$ and $\tilde{P}_i$ denote latent (pre-truncation) values. Initialize:

$$\tilde{C}_1 = \mu_1^C, \qquad \tilde{P}_1 = \mu_1^P. \tag{5}$$

For $i \geq 2$,

$$\tilde{C}_i = \mu_i^C + \rho_C\big(\tilde{C}_{i-1} - \mu_{i-1}^C\big) + \varepsilon_i^C, \tag{6}$$

$$\tilde{P}_i = \mu_i^P + \rho_P\big(\tilde{P}_{i-1} - \mu_{i-1}^P\big) + \varepsilon_i^P, \tag{7}$$

where $\rho_C, \rho_P \in [0, 1)$, and $\varepsilon_i^C \sim \mathcal{N}(0, \sigma_c^2)$, $\varepsilon_i^P \sim \mathcal{N}(0, \sigma_p^2)$, truncated below.

Observed values enforce non-negativity:

$$C_i = \max(0, \tilde{C}_i), \qquad P_i = \max(0, \tilde{P}_i). \tag{8}$$

3.6 Stopping Time and Sleep Latency

The stopping time is

$$S = \min\{i \geq 1 : C_i \leq \delta\}. \tag{9}$$

Total time to sleep:

$$T_{\text{sleep}} = \sum_{i=1}^{S-1}(C_i + P_i) + C_S. \tag{10}$$

Note that P_S is not added, since the infant is asleep during or immediately after the terminal cry.

4 Simulation Methodology

The model is implemented in Python 3. Parameters are summarised in Table 1. For each parameter set, we run $N = 1000$ Monte Carlo simulations and record the distribution of T_{sleep}.

Table 1: Parameter summary and baseline values.

Parameter	Meaning	Domain	Baseline
C_1	Initial cry duration	> 0	7 s
P_1	Initial pause duration	> 0	10 s
α_0	Baseline cry decay	> 0	0.30
α_1	Circadian amplitude (α)	$[0, \alpha_0)$	0.10
γ_0	Baseline pause growth	> 0	0.40
γ_1	Circadian amplitude (γ)	$[0, \gamma_0)$	0.10
β	Max additional pause	> 0	15 s
ϕ	Circadian phase	$[0, 24)$	21 h
ρ_C	Cry autocorrelation	$[0, 1)$	0.30
ρ_P	Pause autocorrelation	$[0, 1)$	0.30
σ_c	Cry noise SD	> 0	0.50
σ_p	Pause noise SD	> 0	1.00
δ	Sleep threshold	> 0	1 s
$N_{\max}$	Max cycles	$\mathbb{N}$	50

5 Results

5.1 Single Trajectory

Figure 1 shows one simulated trajectory. Cry durations decay exponentially, while pauses grow logistically until the sleep threshold is reached.

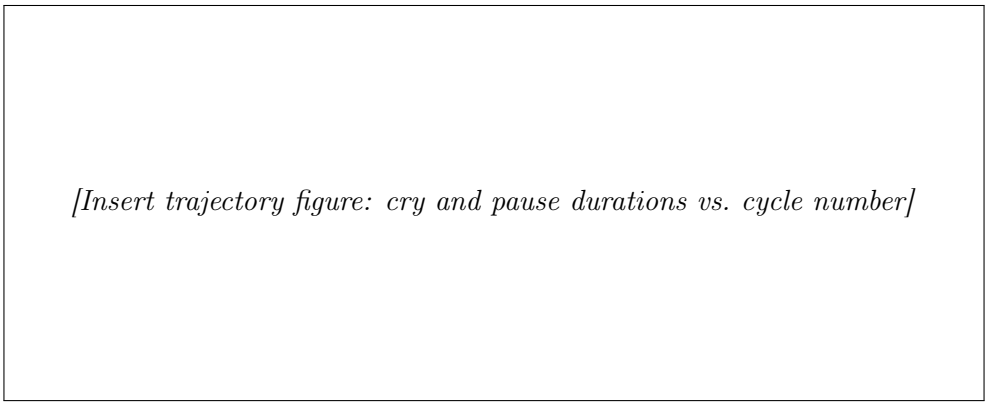


Figure 1: One simulated cry–pause trajectory. Cry durations (circles) decay; pause durations (squares) saturate. Sleep occurs when the cry falls to δ.

5.2 Distribution of Sleep Onset Time

Figure 2 shows the Monte Carlo distribution of sleep onset time when starting at $t_0 = 21{:}00$.

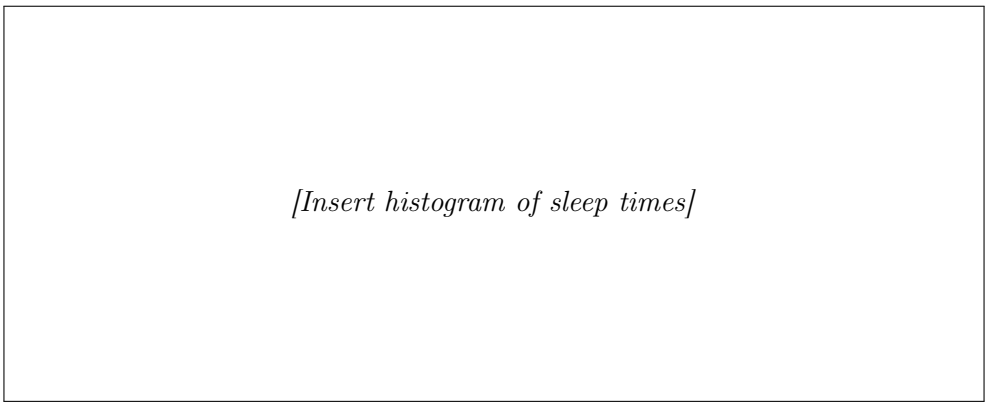


Figure 2: Monte Carlo distribution of sleep onset time for $t_0 = 21{:}00$ ($N = 1000$ simulations).

Summary statistics (placeholder):

- Mean sleep onset: **[MEAN TIME]**
- Median: **[MEDIAN TIME]**
- 95% prediction interval: **[LOWER]**–**[UPPER]**
- Mean number of cycles: **[MEAN CYCLES]** (range: **[MIN]**–**[MAX]**)

5.3 Sensitivity Analysis

Figure 3 varies the cry decay rate α_0 and reports the resulting mean sleep latency. Larger α_0 produces shorter latency, as expected.

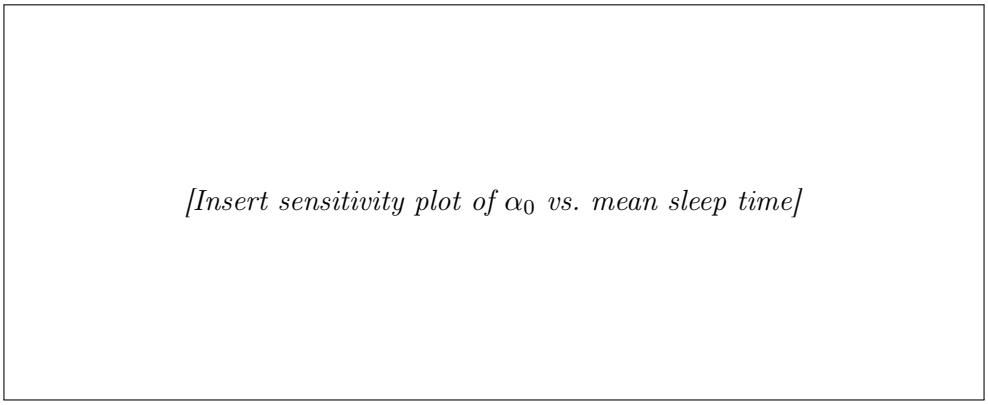


Figure 3: Sensitivity of mean sleep latency to α_0, holding other parameters fixed.

6 Discussion

6.1 Interpretation

The model reproduces qualitative features of observed infant sleep onset: cycles converge, pauses saturate, and sleep latency is bounded. The circadian modulation ensures that night-time crying resolves faster, consistent with elevated homeostatic sleep pressure.

6.2 Limitations

We acknowledge the following limitations:

1. **Truncated Gaussian noise.** Cry and pause durations are non-negative and right-skewed; Gamma or log-normal noise may be more appropriate.
2. **No external interventions.** Real caregiving includes soothing, feeding, and diaper changes.
3. **Fixed parameters per infant.** State changes (hunger, illness) are not modelled.
4. **No empirical calibration.** Parameters are illustrative, not estimated from data.
5. **Sinusoidal circadian assumption.** Real circadian rhythms may not be perfectly sinusoidal.

6.3 Ethical Considerations

This model is intended for *retrospective analysis* and *expectation management*, not as a substitute for parental attention. Parents and caregivers should always respond promptly to a crying infant. Prolonged unresponsive crying is associated with elevated infant cortisol and, in extreme cases, attachment insecurity. The model does not and cannot replace clinical assessment, common sense, or the caregiver's instinct.

7 Future Work

We plan three extensions.

(1) Empirical calibration. Collect audio recordings of infant crying and fit the model via maximum likelihood or Bayesian inference.

(2) Bayesian hierarchical modelling. Introduce priors on $\alpha_0, \beta, \gamma_0$ and allow partial pooling across infants. This yields full posterior predictive distributions and quantifies inter-infant heterogeneity. Implementing this in `PyMC` is straightforward.

(3) Covariate extension. Include feeding time, room temperature, and age as predictors of α and γ. This turns the model into a regression framework suitable for clinical use.

8 Conclusion

We have proposed a stochastic model of infant cry–pause dynamics and derived its sleep-onset distribution from first principles. The model is simple, interpretable, and extensible. It offers a statistically principled language for describing a phenomenon that has, until now, been discussed mainly in qualitative terms.

Code Availability

All simulation code is available at `https://github.com/[YOUR-USERNAME]/babycrying`.

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